Machine learning in battery management systems: a two-tier scoping review of 799 eligible studies with in-depth classification and validation audit

Hussain Touhid Siddiquee, Syeda Salsabil Islam Ariya, and Jasimul Islam Chowdhury

Abstract—Machine learning (ML) has moved from a promising addition to battery research to a load-bearing component of battery management systems (BMS), but the validation practices behind the accuracy claims are rarely examined. This paper presents a structured scoping review with a two-tier evidence base: an eligible corpus of 799 peer-reviewed journal articles admitted by uniform rule-based criteria from a capped 30-query Crossref retrieval (798 dated 2020–2026 plus one early-access record stamped 2027), and an in-depth classified subset of 103 studies — 98 from that pool plus five individually verified seed studies, one dated 2019 — coded by function, method family, chemistry, data source, and validation practice. Union-share bibliometrics across the eligible corpus locate the center of gravity of the screened literature in health estimation (SOH), state estimation (SOC), and prognostics (RUL) — query-tag shares, read for ordering rather than exact rates — with charging control, cell balancing, and vehicle- or fleet-level work comparatively under-served for a transportation-electrification readership. Within the classified subset, three weaknesses recur. Only 8 of 88 primary-application papers evaluate online or on real vehicles; 17 of 88 rely on within-cell or k-fold splits that can leak trajectory information; and the largest single data block is private laboratory data (42 of 88: 36 own-cell cycling, 6 laboratory impedance), a reproducibility constraint at least as limiting as the benchmark concentration (23 papers on three datasets). Validation practice shows no detectable difference between physics-informed/hybrid and purely data-driven papers under either definition of stringency — an underpowered contrast in which meaningful differences in either direction cannot be excluded — and symmetric and strong-to-weak label-noise scenarios bracket the strong-protocol share between 11% and 59%; the strong-to-weak floor is a worst-case bound rather than an estimate. A post-hoc recall audit against the included reviews' reference lists bounds what the capped retrieval misses. The paper closes with a requirement-level mapping of BMS functions onto UN Global Technical Regulation (GTR) 20, GB 38031, and ISO 26262, a positioning against prior reviews, and a twelve-item reporting checklist; all data and code are released.

Index Terms—machine learning, battery management systems, lithium-ion batteries, state of charge, state of health, remaining useful life, deep learning, physics-informed neural networks, transfer learning, battery safety

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The authors are with Leading University, Sylhet, Bangladesh. The complete evidentiary record of this review—classified corpus, screening provenance log, protocol audit, codebook, and the blind inter-coder reliability study, together with the full analysis pipeline—is released as a Zenodo deposit, DOI: 10.5281/zenodo.23436202; see Data and Code Availability. Item 12 of the reporting checklist this paper proposes makes such a DOI-backed deposit the reporting norm.

I. Introduction

A. The battery management problem

Lithium-ion batteries now anchor three converging transitions — electrified road transport, renewable-heavy stationary storage, and the electrification of aerospace and marine systems — and in each the battery is an actively managed asset whose usable capacity, lifetime, and safety depend on real-time operating decisions. The battery management system (BMS) is the embedded cyber-physical layer that makes those decisions — sensing, state estimation, protection, balancing, and charging negotiation [1]. The difficulty stems from the fact that the quantities that matter most — state of charge (SOC), state of health (SOH), internal temperature, and the onset of degradation or fault mechanisms — are not directly measurable and must be inferred from noisy, partially observable terminal measurements under widely varying conditions. Model-based approaches built on equivalent-circuit or electrochemical models face a persistent fidelity-versus-tractability trade-off, and their parameters drift as cells age [2]. It is in this inference-and-decision layer that machine learning has made its most forceful entry, and the stakes are highest in transportation electrification: electric-vehicle packs face automotive safety expectations — UN Global Technical Regulation (GTR) 20 on electric-vehicle safety (in-vehicle rechargeable electrical energy storage systems), GB 38031 on traction battery safety requirements, and ISO 26262 on functional safety of electric/electronic systems — that make state estimation, degradation prognosis, and fault detection safety-relevant BMS functions, not research add-ons. Section IX returns to these three standards and maps each estimation function to the property the standard actually demands (detection reliability, gradient/temperature limits, protection timing) — the mapping that turns the validation findings of this review into vehicle-level stakes.

Machine learning entered battery management through estimation on hand-engineered features. Two studies then reset expectations. Severson and colleagues predicted the entire cycle life of lithium-ion cells with a median test error near 9% from the first hundred cycles alone, on a public dataset of 124 cells [3]. Attia and colleagues demonstrated closed-loop Bayesian optimization of fast-charging protocols, identifying high-cycle-life protocols among 224 candidates in 16 days of cycling rather than the

more than 500 days an exhaustive search would take [4]. The five years that followed produced applications across every BMS function: recurrent and convolutional networks for SOC and SOH [5] [6], graph and transformer architectures for prognostics [7] [8], physics-informed neural networks (PINNs) embedding electrochemical and thermal priors [9] [10], transfer learning across chemistries and vehicles [11] [12], reinforcement learning agents [13], and, most recently, large language model (LLM)-based studies of station-side incident-risk analysis and vehicle-cloud management [14] [15].

B. Existing reviews and the literature gap

The review literature has grown alongside the applications but is fragmented along the same axes as the primary studies: an early perspective on data-driven SOC and SOH prediction [16] set the agenda; function-specific reviews cover SOC estimation [17] [18], SOH estimation [19], and prognostics [20]; method-specific reviews follow deep-learning state estimation [1], physics-informed impedance diagnostics [21], or interpretable prognosis [22]; and domain reviews examine fault diagnosis [23] [24] and thermal management [25], alongside broader degradation-method surveys [26] and EV-powertrain ML reviews with BMS scope [27]. What is missing is a cross-functional account of the field's trajectory: which method families are used for which tasks, on which data, validated how — and, among existing cross-functional surveys, none to our knowledge quantifies validation practice at all. Table VII in Section IX positions this review against its closest competitors and states the added value explicitly; in brief, prior reviews synthesize what is claimed, while this review additionally measures — within a 103-study classified subset — how those claims are validated, ties both to three safety standards, and releases the record-level dataset to make every count auditable.

C. Contribution of this paper

This paper makes four contributions: (i) a rule-based eligible corpus of 799 records (798 dated 2020–2026 plus one early-access record stamped 2027), selected by uniform criteria applied to all screened Crossref records — no function quotas, year caps, or citation thresholds — so that field-level shares are measured, not curated, with the retrieval, screening, and tagging provenance of every record released; (ii) an in-depth classified subset of 103 studies from that corpus, coded along the six classification fields (function, method family, chemistry, data source, validation practice, and coded finding/limitation summaries) with the classification dataset, codebooks, per-paper protocol audit, and a two-round 30-paper re-coding protocol — a blind abstract-level round (three of five fields $\kappa \geq 0.88$) followed by a revised-codebook rule-application round on a shared evidence base ($\kappa = 0.92$–1.00) — released together with a label-noise sensitivity analysis of the headline counts; (iii) a two-tier synthesis — measured bibliometrics from the eligible corpus (Section III) plus function-by-function critical review of the classified subset (Sections IV–VIII) — with a BMS-function-to-safety-standard mapping and a gap-driven research agenda including a twelve-item reporting checklist (Section IX); and (iv) a transportation-electrification cut of the corpus (Section III-B) separating cell-level laboratory evidence from pack-, vehicle-, and fleet-level practice. Fig. 1 previews the 88 primary studies of the classified subset, mapping them across management functions and method families: the center of gravity sits in the health-prognostics complex, hybrid model–data approaches span every function, and learning-based charging control remains a concentrated Bayesian-optimization stronghold. The remainder of the paper is organized as follows. Section II details the review methodology and its limitations. Section III presents the two-tier corpus overview, five-year bibliometrics, and the transportation-electrification cut. Sections IV–VII review machine learning for state estimation, health estimation and prognostics, charging control, and thermal management, faults, and safety. Section VIII examines cross-cutting directions, and Sections IX and X give the research agenda and conclusions.

II. Review Methodology

A. Search strategy

All candidate records were retrieved programmatically from the Crossref REST API between 4 and 5 September 2026 using 30 bibliographic query strings spanning the application areas above. Queries were issued in five passes: a citation-sorted 2020–2026 pass, dedicated 2020–2022 and 2023 windows, a mid-window 2021–2023 coverage pass, and a relevance-sorted recency pass starting June 2024, yielding 3,000 fetched rows; document-type, year, and title-completeness filters left 2,973, and DOI-based deduplication (1,423 duplicates removed) left 1,550 unique records. Six landmark studies — including [3] and [2] — were additionally safeguarded as forced seeds: five were retrieved and verified individually because citation-sorted retrieval had crowded them out, and one was already among the top-cited hits. Four properties should be stated plainly. First, the Crossref registry is a metadata registry, not a discovery engine like Scopus or Web of Science; it was chosen so that every included record is individually verifiable by direct DOI resolution, at the cost of a recall floor. Second, citation-ordered passes favor established English-language work; all trend statements therefore describe this citation-weighted, journal-article sample rather than the field in the aggregate. Third, January–May 2024 is a known retrieval hole — the citation-sorted pass returns almost nothing from those months and the recency pass starts at June 2024 — so the 2024 dip visible in year distributions is a design artifact of this retrieval, not evidence about publication volume; per-year counts in Section III-A are therefore reported as counts within the capped retrieval, with the hole documented, and no field-level per-year shares are claimed. Fourth, every query pass returned its top 20

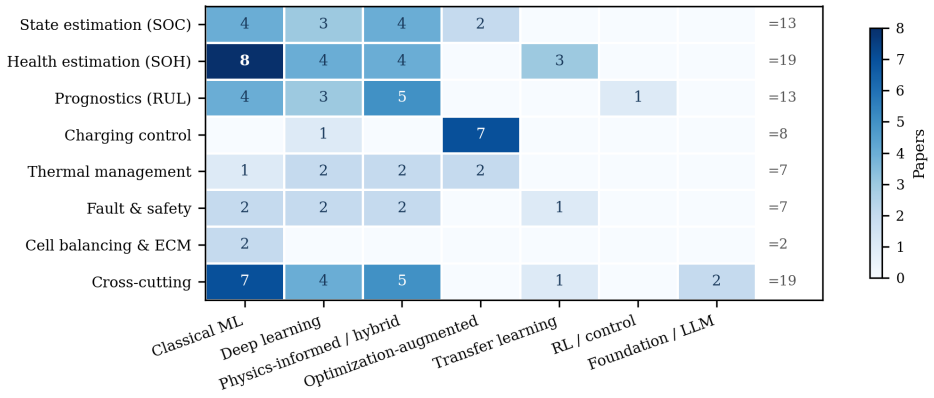


Fig. 1. The classified subset mapped across battery-management functions and method families (the 88 primary-application papers). The remaining 15 classified papers are 12 method-coded reviews ("Review / perspective") plus 3 review-type syntheses coded under their dominant method family (two physics-informed, one classical), all 15 of which are counted in the Reviews column of Table II. Rates that use the primary denominator are 88, not 103.

records only, so the 30 queries are design-capped at 100 records per query family; a post-hoc recall audit against the deposited reference lists of the 15 included reviews and six seed studies finds that the retrieval contains 112 of the 649 in-scope-looking, 2020-onward, DOI-bearing references they cite (17.3% counted per-source; 96 of 586 unique references, construction and limits in the supplement S4; construction and limits in the supplement S4; construction and limits in the supplement S4; 16.4%). Retrieval recall is therefore bounded and now quantified: theme frequencies, per-family shares, and 'earliest' claims in this review are scoped to this retrieval and the subset it fed, and the LLM-estimator recall miss disclosed in the cross-cutting section is cited by none of those reviews either (audit released as recall_audit.py/.json with cached responses). A released retrieval-depth pilot quantifies what the cap withholds: re-running six of the 30 query families with cursor paging to ten times the per-pass depth, under the original windows and sort orders, yields 1,500 title-gated records absent from the 1,550 (1,406 venue-eligible) — implying an all-family net gain on the order of 5,000 eligible records at the original design's observed overlap factor. The cap, not the registry, is the binding constraint; lifting it is the pipeline's single highest-leverage upgrade (Section IX-C, direction (g)), and the corpus is read as a capped sample throughout.

B. Screening and rule-based selection

The screening funnel is shown in Fig. 2. A title-level relevance gate requiring both battery-related and machine-learning terminology retained 883 records and excluded 667 (predominantly adjacent-field papers that surfaced through loose matching); the gate is a mechanical keyword rule, and its term lists are released (tagging_terms.json) together with all 1,555 screening-log rows (the 1,550 retrieved records plus the five individually retrieved seeds; screening_log.csv). Venue and document-type screening removed 84 records — conference abstracts and proceedings series, preprint servers, and non-scholarly venues — leaving 799 eligible records. Selection proceeds in two tiers. Tier 1 (eligible corpus). Every one of the 799 eligible records is included, subject only to venue screening; no function quotas, year caps, or citation thresholds are applied. Every one of the 799 eligible records carries a Crossref-registered DOI, and every record carried forward to Tier 2 full-text coding was individually DOI-resolved. Function tagging at this tier is inherited from retrieval: each record carries the area of every query theme that returned it (query list released), so a record can hold several tags; 202 of 799 records (25%) carry no management-function tag — most are method-level or cross-cutting records (physics-informed, second life, digital twins, benchmark papers) — so per-function shares below are computed over tagged records and cross-function totals are reported as union shares, never as sums of per-tag percentages. Tier 2 (classified subset). Because in-depth coding of chemistry, data source, and validation practice requires full-text assessment, full-text coding was applied to a classified subset of 103 studies — 98 from the eligible pool plus 5 landmark seeds retrieved and verified individually — curated with the same eight-function stratification used by the tagging pass, with six landmark studies (including the 2019 field-defining seed, Severson et al.) safeguarded as forced entries — five retrieved individually from outside the query pool and one already among the top-cited hits (Section II-A). This subset is curated, not sampled: the released record-level log identifies every record at every screening stage (with the forced seeds annotated), the released selection files allow every inclusion and drop to be traced, and its function and year composition does not match the eligible corpus's measured shares exactly. Section III therefore never generalizes subset distributions to the field; where the subset is the only source (method family, validation practice), the text reports subset statistics and their selection caveats explicitly.

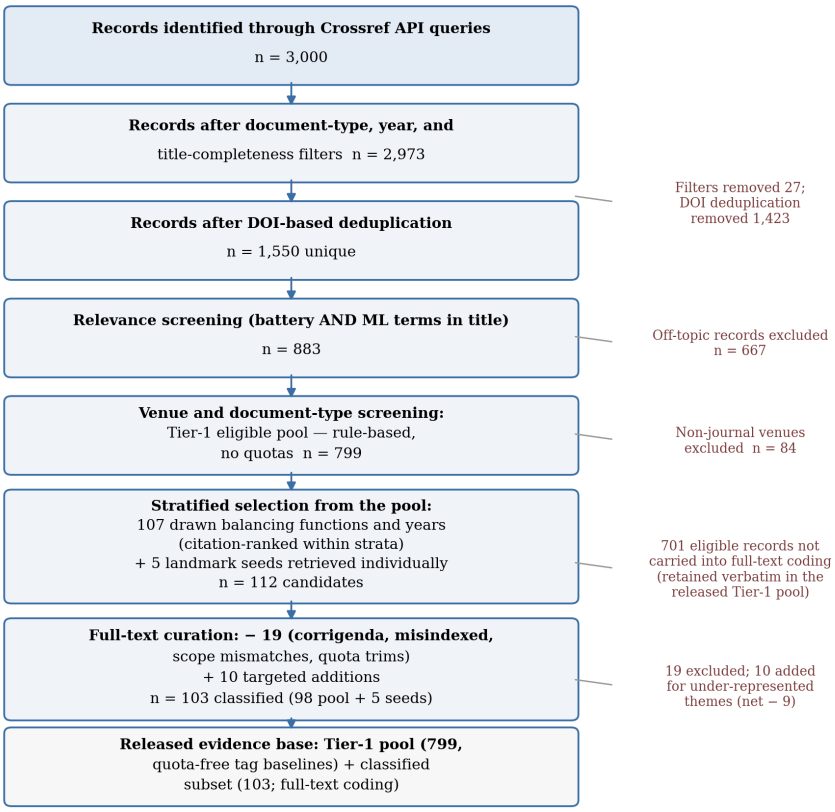


Fig. 2. Study-selection flowchart with the two-tier design. Records were retrieved from the Crossref REST API on 4–5 September 2026 (five passes; January–May 2024 retrieval hole documented in Section II-A), deduplicated, screened at title level by a released mechanical keyword rule (883 retained, 667 excluded), then at venue and document-type level. Tier 1 includes every eligible record (799, rule-based, no quotas); Tier 2 carries a curated full-text-classified subset of 103 with six landmark studies safeguarded (one already among the top-cited hits; five force-included individually from outside the pool. Record-level provenance is released as screening_log.csv (all 1,555 rows: 1,550 retrieved records plus the five individually retrieved seeds) and tagging_terms.json.

C. Classification framework, validation definitions, and limitations

Each paper was classified along six fields: management function (eight values, including a cross-cutting cluster); dominant method family (classical ML, deep learning, physics-informed/hybrid, optimization-augmented, reinforcement/control learning, transfer learning, foundation models/LLMs, review); chemistry; primary data source; validation approach; and a coded summary of the headline finding and principal limitation. Three terms are used consistently: an eligible record is one of the 799 Tier-1 studies; a classifiable paper is one of the 103 Tier-2 studies; and a primary-application paper is one of the 88 classified papers whose contribution is a primary application — the 12 method-coded reviews and 3 review-type syntheses coded under their dominant method family are excluded from every rate that uses this denominator (Table II counts them per function; papers.csv identifies each one).

Validation practice is coded against four definitions used throughout this paper: (1) within-cell split — train and test segments drawn from the same cycling trajectory; (2) unsupervised k-fold cross-validation — folds assigned over samples of the same trajectories without cell isolation; (3) cross-cell holdout — every test cell held out entirely from training (with cross-chemistry and cross-domain holdout extending this isolation across chemistry or application); and (4) online or real-vehicle/fleet validation — the deployed system evaluated on live vehicle data. Cross-cell, cross-chemistry, cross-domain, and online/fleet are together the strong protocols; the four remaining coded environments (experimental hardware, simulation plus hardware spot-check, simulation-only, case study) characterize where a study was exercised but say nothing about train–test isolation, and they are reported separately — never mixed into strong-protocol percentages. Borderline assignments, the absorbing rule for ambiguous cases (a study is credited with the strongest protocol it reports for its headline claim), and the decision tree with examples are given in the released codebook. Both leaking definitions share one mechanism: samples adjacent in time on the same trajectory are strongly correlated, so a split that places segments of one cell on both sides of the train–test boundary lets a model score well by memorizing cell-specific trajectory structure rather than by learning the cross-cell dynamics that onboard use on an unseen pack would require.

Classification was performed by the lead author with

conservative coding from full texts; to probe the reliability of that coding, two co-author coders (the second and third authors, not involved in the primary coding) blind-re-coded a 30-paper stratified random sample (drawn by function and year band, seed-pinned and reproducible) under the released codebook, without access to the released labels, to the population statistics, or to each other. The blinding was procedural rather than external: the coders are authors of this study — their later author-contribution roles included verification of classifications and counts — but their coding sheets were locked and hash-tracked at completion, with a filed blinding memo documenting what each coder had opened before coding. Blind agreement between the two coders is strong on three of the five classification fields: management function (κ = 0.92, bootstrap 95% CI 0.80–1.00; Krippendorff $\alpha = 0.92$), method family ($\kappa = 0.88$, CI 0.73–1.00; $\alpha = 0.88$), and primary data source ($\kappa = 0.88$, CI 0.72–1.00; $\alpha = 0.88$). Two fields fell below the $\kappa \geq 0.66$ threshold: chemistry (κ = 0.55, CI 0.16–0.83) and validation protocol ($\kappa = 0.48$, CI 0.31–0.65). Part of the shortfall is an evidence-access artifact: the two coders had full-text (or preprint) access to only 7 and 2 of the 30 sample papers, respectively, so 23 and 28 were coded from abstracts alone — and these two fields are precisely those least often stated at abstract level. The consequence extends to the aggregation underlying Section III's counts: on strong-protocol versus non-strong, the two coders agree on only 25 of 30 papers (raw agreement 83%, $\kappa = 0.52$ at the sample's prevalence, bootstrap 95% CI 0.09–0.84 — below the 0.66 threshold applied throughout this paper). The released labels are therefore treated as one plausible coding, and all headline validation counts in Section III carry a scenario sensitivity analysis built from the observed coder-versus-label disagreement in this sample: the central strong-protocol estimate (48% of the 88 primary papers) spans 40–59% under the symmetric scenario, and a strong-to-weak worst-case bound (Section III-C) places it as low as 18% at the observed abstract-level recode rate — so the strong-protocol level is a bounded range rather than a pinned estimate, and no qualitative conclusion below depends on where in these ranges the truth lies (Section III-C). Coder access constraints, training, and the complete per-paper, per-field disagreement table are released together with both locked coding sheets and the hash-tracked log. The protocol's second stage for the below-threshold fields — codebook revision, then one re-coded subset — was then executed as pre-declared. The codebook was revised to v1.1, codifying a not-ML method code, a production-and-quality area code, a field-stationary data code with a self-lab/EIS-lab tie-break (EIS: electrochemical impedance spectroscopy), the dedicated "not reported" validation code replacing the divergent weak-code fallbacks that had driven the low round-1 agreement on this field, and a fixed reading for cross-function reviews; the chemistry inference-rule candidate remains logged but not codified. Full texts were then sought for all 30 sample papers (10 obtained, plus one accepted-manuscript preprint and the released code repository of one SOH study; 19 remain abstract-only behind paywalls), and both coders re-coded the same sample from that shared evidence base in a single session, so the round-2 κ values measure residual rule ambiguity of the revised codebook rather than coder independence; because the v1.1 patches were derived from this same sample's round-1 disputes and agreement was re-measured on that sample, the round-2 design is same-sample by construction and is reported as a rule-application check, not independent dual coding; the two rounds are complementary rather than interchangeable. All five fields then pass the 0.66 threshold at the 0.80 strong level — management function $\kappa = 0.96$, method family 1.00, chemistry 1.00, primary data source 0.96, validation protocol 0.92 — with the strong/non-strong bucket rising from 25 of 30 papers (83%, $\kappa = 0.52$) to 28 of 30 (93%, $\kappa = 0.83$). Four residual disagreement cells are released with the sheets; the full-text reads surfaced five candidate label corrections (among them two Severson-dataset papers whose released chemistry records NMC where the dataset itself is LFP, and one study whose released labels differ from its own released code repository on all five fields), itemized with evidence, left unapplied, and queued for author adjudication.

The synthesis proceeds on two tracks: measured bibliometrics from the eligible corpus and cross-tabulations from the classified subset, both generated programmatically from the released datasets (Section III), and qualitative function-by-function critique (Sections IV–VIII). The methodology's limitations are acknowledged: Crossref coverage excludes influential conference papers (a descriptive conference-venue probe in the supplement, S6, finds no evidence that this journal-only design understates stated validation evidence, but at 14 papers from six venues it cannot support inference); title-level screening can miss relevant work; citation-ordered retrieval privileges established English-language venues; the per-year retrieval window has one documented hole (Section II-A); the Tier-2 subset is curated rather than sampled, so validation-practice rates are subset statistics that do not automatically generalize to the eligible corpus or the field; and reliability is documented on a blind, two-round 30-paper re-coding subset (two co-author coders — three of five fields above threshold in round 1, all five at the strong level in the revised-codebook round 2, whose sheets were produced in a single session and which measures rule-application agreement on a shared evidence base rather than independent full-text dual coding) rather than measured corpus-wide, with the resulting sensitivity ranges reported; the strong/weak taxonomy is itself design-only — it codes the split architecture, not how many cells were held out, whether tuning touched the test set, or whether preprocessing was fit before splitting — so a strong code certifies the design of a validation, not its execution. DOI resolution, moreover, verifies that a record exists, not that its venue is topically sensible; the venue screen is therefore document-type-based rather than topical, and the released audit file (final_audit.json) records each

reference's Crossref-resolved title and venue at retrieval time so that any such case is visible rather than hidden. These are the reasons this paper describes itself as a scoping review with bibliometric classification rather than a meta-analysis.

III. Corpus Overview and Five-Year Evolution

A. Publication trend and method-family evolution

Two layers of trend evidence are reported. On the measured tier (eligible corpus, 799 records), the capped retrieval captures 55, 85, 139, and 223 journal articles for 2020–2023, 56 for 2024 — an observed count inside the capped retrieval, not an estimate of missing volume; the January–May 2024 window is only partially covered (Section II-A) — then 126 in 2025 and — partial year, retrieval date September 2026 — 114 in 2026, plus one further early-access record date-stamped 2027 by Crossref metadata and omitted from the year chart (Fig. 3, left panel). These are counts within a capped retrieval, not census totals, and they do not support absolute-growth claims ("the literature quadrupled" would require an uncapped retrieval). A per-year external denominator was tested and discarded: Crossref's reported result totals for the same 30 queries (a published script and its output ship with the deposit) rely on a loose token-level bibliographic match whose totals run into the hundreds of thousands and vary strongly with query phrasing, so they are not interpretable as field denominators. What remains defensible is direction within the sample: the eligible-corpus counts rise through 2020–2023, dip at the retrieval hole, and remain high in 2025–2026, and the tagged composition of those records shifts toward estimation and hybrid methods (next paragraph).

On the classified subset, the compositional shift in method families (Fig. 3, right panel; Table I) is a within-subset sequence, not a measured field trend. Classical ML — Gaussian process regression, support vector machines, ensembles, shallow networks on engineered features — contributes 12 of the 23 subset papers published from 2019 through 2021, while physics-informed and hybrid model–data learning rises year by year (1, 2, 3, 3, 4, 4, and 5 primary papers in 2020–2026) — the only multi-year family with a monotonically non-decreasing trajectory within the subset — so that 13 of its 22 primary papers fall in 2024–2026 and it is the largest single family in 2026 (Table II gives the subset totals by year). Optimization-augmented learning (Bayesian optimization, surrogate-assisted and meta-heuristic schemes) peaks in 2024 (5 of its 11 papers); transfer learning accelerates in 2024–2025 (3 of its 5 papers); reinforcement learning appears as the dominant family in 1 paper and as an embedded component in at least one more [13] [28]; the subset's first foundation-model and LLM-based studies appear in 2026 (2 papers) [14] [15], with journal-published LLM-based battery estimation beginning in 2025 outside this retrieval, as a retrieval miss rather than a venue exclusion [29]. These family shares describe the subset only; the subset is curated, small-denominator years carry wide sampling error, and the eligible pool's method-theme tags show no monotonic physics-informed trend. Two qualifications matter: the families are not mutually exclusive (a particle swarm optimization (PSO)-tuned long short-term memory (LSTM) is both deep learning and optimization-augmented), so the composition understates method mixing; and physics-informed papers show no detectably different validation stringency from the others even under the most permissive definition — a nominally lower strong-protocol rate that these counts are underpowered to resolve (Section III-C).

B. Measured application areas, the transportation-electrification cut, and venues

Because function tags were applied mechanically to all 799 eligible records, the application-area distribution is measured rather than quoted from a curated sample. Tags are inherited from the retrieving queries (a record carries every area whose query returned it); 597 records (75%) carry at least one management-function tag, while the remaining 202 are method-, second-life-, digital-twin-, benchmark-level, or unmapped records without a specific BMS function, and the tags match the subset's full-text function labels for 53 of the 75 core-area papers found in the pool (71%, any-tag match — a lenient criterion, so shares are read for ordering, not for exact rates). Of all eligible records, health estimation (SOH) leads at 20% (158 records; 27% of those function-tagged), followed by state estimation (SOC) at 17% (133), prognostics (RUL) at 13% (103), thermal management at 12% (99), fault and safety at 12% (96), charging control at 8% (64), and cell balancing at 5% (38). These per-tag percentages must not be summed — a record can carry several tags; the union of SOC, SOH, and RUL covers 46% of eligible records, and adding the four control-and-safety functions brings management-function coverage to 75%. The estimation concentration is a measured pattern consistent with the commercial pull of lifetime prediction for warranties and second-life markets, and is mirrored (with a heavier cross-cutting share) in the classified subset (Fig. 1; Table II gives subset composition by year; Table IV maps representative studies). Two measured concentrations matter specifically for transportation electrification. First, the electrification cut: only ≈2% of eligible records (n = 17) are tagged fleet- or cloud-orchestration-adjacent, and in the classified subset only 10 of 88 primary papers use real vehicle or fleet data as their primary source against 36 resting on laboratory cells — the classified literature is predominantly cell-level and laboratory-scale, one integration level below the pack- and vehicle-level questions that UN GTR 20, GB 38031, and ISO 26262 pose. Second, venue concentration is marked: in the eligible corpus the Journal of Energy Storage (103 records), Energy (61), and Batteries (45) lead, and the same two energy-storage journals head the classified subset (Journal of Energy Storage with 15 papers and Energy with 13, Applied Energy third at 9) — energy-storage venues, not transportation-electrification venues,

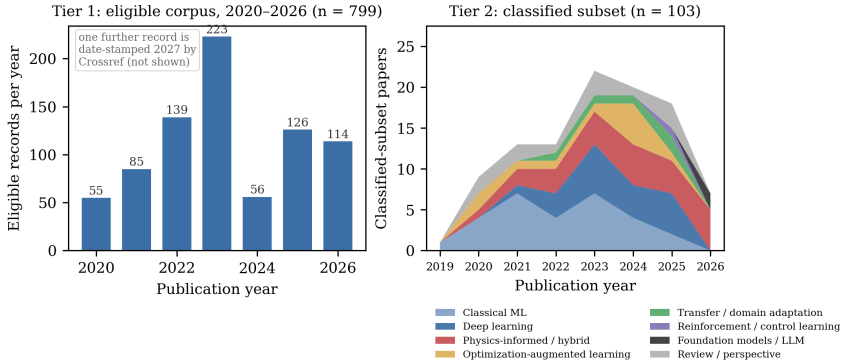


Fig. 3. Two-tier trend evidence. Left: Tier 1 eligible-corpus records per year across the capped retrieval (the 2024 dip is the documented January–May 2024 retrieval hole, not a decline in the field; one further record is date-stamped 2027 by Crossref and not shown). Right: Tier 2 classified-subset papers by year and method family, with reviews shown as their own band; the yearly totals match Table II, and the three review-type syntheses sit inside their dominant family bands (two of them physics-informed), so the physics-informed band exceeds the primary-only trajectory stated in the text (curated subset; wide sampling error in small years).

TABLE I

Method families: representative techniques, strengths, and limitations. Counts are the 88 primary-application papers; the three review-type syntheses coded under a method family plus the 12 method-family reviews are accounted for in the function-by-year composition table. Abbreviations: SVM, support vector machine; SVR, support vector regression; LSTM, long short-term memory; GRU, gated recurrent unit; CNN, convolutional neural network.

Method family	Papers	Representative techniques	Strengths	Limitations
Classical ML	28	Gaussian process regression, SVM/SVR, ensembles, shallow networks on engineered features	Native predictive distributions (calibration must be verified empirically); small-data friendly; cheap to embed	Feature engineering is manual; ceilings under complex dynamics
Deep learning	19	LSTM/GRU, CNN, attention/Transformer, autoencoders, graph networks	Learns features from raw curves; strong trajectory modeling	Data-hungry; weak extrapolation; opaque
Physics-informed / hybrid	22	PINNs, physics-constrained autoencoders, filter-embedded networks, multiphysics fusion	Structure aids extrapolation; reduced data demand	Inherits simplified-model bias; harder to train
Optimization-augmented	11	Bayesian optimization, surrogate-assisted multi-objective design, metaheuristic-tuned networks	Sample-efficient protocol discovery; explicit trade-off handling	Offline tuning cost; surrogate bias
Transfer / domain adaptation	5	Domain-adversarial training, fine-tuning across chemistry/protocol	Directly targets the generalization gap	Negative transfer; needs shared protocols
Reinforcement / control learning	1	Deep-RL prognostic model selection (also an embedded component in further studies)	Adaptive decision-making along trajectories	Training stability; scarce hardware validation
Foundation models / LLM	2	LLM incident-risk analysis; LLM-based SOH estimation with uncertainty-gated physics fallback	Reasoning over heterogeneous text/system context; transferable priors	Hallucination risk; latency and cost
Review / perspective	12	Narrative and systematic syntheses	Maps the field; identifies shared gaps	No primary validation

absorb the field's output. The thin representation of cell balancing is also telling: balancing remains dominated by classical control practice, and its most visible result in this subset is a single contrarian study reporting that routine balancing can be dispensed with in well-managed packs, reducing hazardous lithium-plating side reactions [30] — reproduced here as one study's claim, not an established result.

C. Data landscape and validation practice

The data underpinning the classified subset are strikingly concentrated: 36 of the 88 primary-application papers rest on laboratory cycling of the authors' own cells; the public Severson/Toyota–MIT–Stanford (MATR) dataset alone supports 11, NASA and CALCE 6 each, and laboratory impedance spectroscopy 6; 12 are simulation-based. Only 10 papers use real vehicle or fleet data as their primary source, and data use is a distinct measure from validation protocol: 8 of the 10 fleet-data papers are validated online or on real vehicles, and the other two through cross-domain holdout or case-study protocols (Table III). Validation practice is arguably the classified subset's most consequential weakness, and it is now defined and audited explicitly. Per the definitions of Section II-C, cross-cell holdout — the strongest common laboratory protocol — accounts for 24 primary-application papers, cross-chemistry holdout for 7, cross-domain holdout for 3, and online or fleet validation for 8; together these strong protocols cover 42 of the 88 primary-application papers (48%). Within-cell splits (12 papers) and unsupervised k-fold cross-validation (5) leak trajectory information

TABLE II
Corpus composition: management function by publication year. The Reviews column counts the 12 method-family reviews plus the 3 review-type syntheses coded under their dominant method family; Primary = Total − Reviews is the 88-paper primary-application count used in Fig. 1.

Function	’19	’20	’21	’22	’23	’24	’25	’26	Total	Reviews	Primary
Cell balancing & ECM	0	0	0	0	1	1	0	0	2	0	2
Charging control	0	2	0	0	2	3	1	0	8	0	8
Cross-cutting	0	1	0	6	8	3	2	4	24	5	19
Fault & safety	0	0	2	0	2	3	2	0	9	2	7
Health estimation (SOH)	0	2	3	2	4	3	4	2	20	1	19
Prognostics (RUL)	1	1	3	2	1	1	6	0	15	2	13
State estimation (SOC)	0	3	4	2	1	5	2	0	17	4	13
Thermal management	0	0	1	1	3	1	1	1	8	1	7
Total	1	9	13	13	22	20	18	7	103	15	88

between train and test, producing optimistic errors that do not survive cell-to-cell variability; the remaining 29 studies establish hardware operation (20), simulation- or case-level evidence without a generalization protocol (9) — 42+17+29 = 88. The complete per-paper protocol assignment is released in papers.csv (valid field) and tabulated as a supplement (protocol_audit.csv), so the 42-of-88 arithmetic above can be verified row by row. The 30-paper blind re-coding of Section II-C is an abstract-level inter-coder agreement check whose sample includes the 5 review-labelled papers — not an independent re-derivation of these assignments (their bucket agreement with the released labels is itself weak, and the revised-codebook round 2 lifts the coders’ mutual bucket agreement to 28 of 30 papers, $\kappa = 0.83$, while remaining an agreement measure rather than a re-derivation, Section II-C); the counts instead carry the scenario sensitivity that follows; a released supplementary note re-derives the sensitivity envelope and the round-2 reliability numbers, keyed to the section numbering of both manuscripts. An independent re-coding of the validation field by the AI research assistant (coder D), performed after the label set was frozen (open-access full texts where retrievable, indexed abstracts otherwise), determinably covers 23 of the 88 primaries: 10 of the 14 determinable strong labels are confirmed, four are contradicted, and two non-strong labels would be upgraded — bucket agreement 17 of 23 (74%), itemized and released with no label altered (corpus/validation_recode/; per-paper evidence in the supplement, S7). An abstract-level baseline screen (supplement S5.2) finds 32 of the 38 abstract-screenable primary papers (84%) naming no model-based or classical baseline, 2 naming one without a same-data comparison, and 4 (11%) stating one — the evidence behind checklist item 6 (Section IX-D). An abstract-level baseline screen (supplement S5.2) finds 32 of the 38 abstract-screenable primary papers (84%) naming no model-based or classical baseline, 2 naming one without a same-data comparison, and 4 (11%) stating one — the evidence behind checklist item 6 (Section IX-D). An abstract-level baseline screen (supplement S5.2) finds 32 of the 38 abstract-screenable primary papers (84%) naming no model-based or classical baseline, 2 naming one without a same-data comparison, and 4 (11%) stating one — the evidence behind checklist item 6 (Section IX-D).

The full cross-tabulation of method family by validation class makes the pattern concrete (Table I gives the family totals). Classical learning accounts for 28 primary papers and uses a strong isolation protocol in 19 of them (68%); deep learning follows at 9 of 19 (47%); physics-informed and hybrid schemes at 8 of 22 (36%); and optimization-augmented learning — Bayesian optimization, surrogate-assisted, and meta-heuristic schemes — at 0 of 11. The optimization-augmented cluster is instead validated by physical demonstration: six of its papers exercise experimental hardware, three rest on simulation, one combines simulation with hardware spot-checks, and one uses k-fold cross-validation. The single reinforcement-learning paper is validated within-cell, and the two foundation-model papers split between one online demonstration and one case study. Transfer learning is the mirror image: all five of its papers validate on isolation protocols (three cross-chemistry, two cross-domain), because its research object is generalization, so its experiments test it. The audit thus exposes a two-culture split in validation practice: the families whose outputs are estimates are exercised on held-out isolation, while the family that actuates the battery is exercised on physical demonstration. The split is consequential for deployment, because the charging and thermal clusters — where learned strategies act directly on the pack — carry no strong isolation protocol at all: 0 of 8 and 0 of 7 primary papers, respectively (Table VI), with demonstration- or simulation-based evidence throughout.

Data sources and chemistry stratify the same evidence by function. The SOC cluster’s 13 primaries rest almost entirely on the authors’ own laboratory data — 12 private-laboratory sources (own cells or laboratory impedance) and one simulation — with no public-benchmark and no real-vehicle source. The SOH/RUL complex of 32 primaries mixes 16 public-benchmark with 11 private-laboratory sources, alongside 4 real-vehicle sources and 1 mixed-source study. Charging (8) splits evenly between private cells and simulation (4 and 4); thermal (7) is simulation-dominated (5 of 7, the other 2 on laboratory hardware); and fault and safety is the only cluster whose modal primary source is real-vehicle data (3 of 7). Chemistry concentrates correspondingly: NMC cells dominate 54 of the 88 primary papers (61%), and the thermal-management primaries are NMC in all seven cases, so

heat-generation and thermal-runaway evidence for other chemistries is thin in this subset. Both matrices can be re-derived from papers.csv, which carries the method, chemistry, data, and validation fields for every classified paper, so each count above is checkable against the record-level data.

Crossing validation practice with method family, the cautious reading is: 8 of the 22 physics-informed/hybrid primary papers (36%) use a strong protocol, against 34 of the 66 other primary papers (52%) — Fisher's exact $p = 0.32$, odds ratio 0.54 with a 95% confidence interval (CI) of 0.21–1.46. Under the broader definition that also counts experimental-hardware demonstration as strong validation, the contrast is 13 of 22 (59%) versus 49 of 66 (74%), OR 0.50 (95% CI 0.19–1.36, $p = 0.19$). We report both and claim neither as a model-family effect: the difference is not statistically significant under either definition, confounding is unadjusted (physics-informed work skews toward early-stage and simulation-heavy studies), and $n = 22$ is small. A minimum-detectable-effect analysis (released with the corpus; supplement S3) shows that 80% power at $\alpha = 0.05$ would require a gap of roughly 30–35 percentage points between the two families, while the observed 15-point gap is detectable with only about 20–25% power — so no detected difference does not mean no difference: meaningful effects in either direction remain compatible with these counts. A label-noise scenario envelope built from the observed blind coder-versus-label disagreement (Section II-C) moves the overall strong-protocol share across 35–52 of the 88 papers (40–59% of the primary papers) under symmetric random flipping and leaves the method-family contrast non-significant in 94% of scenario draws; the observed disagreements are directional and concentrated on strong labels (16 of 19 have the coder defaulting to weak, and each coder recoded 8 of the 13 strong-labelled sample papers weak), so a strong-to-weak scenario — reported as a worst-case bound, because the abstract-level coders defaulted to weak wherever a protocol was not stated — falls to about 16 papers (18%) at the observed recode rate, with an envelope of 10–23 papers (11–26%) and the contrast non-significant in 98% of draws. The method-family contrast is thus robust to plausible coding error in either direction, while the strong-protocol level itself is not pinned by these counts — the worst-case bound says how low it could sit if abstract-level recoding were taken at face value, not where it sits. The noise model is explicit and released (sensitivity_label_noise.py): in each of 4,000 seeded Monte Carlo draws, a strong/non-strong label-flip rate is drawn from a Normal distribution with standard deviation 0.02, truncated at zero — centered on the coders' overall bucket error rate (0.37 and 0.27; mean 0.32) for the symmetric scenario, and on the strong-label-specific recode rate (0.62) for the strong-to-weak scenario — and the drawn rate then flips bucket labels across the 88 primary papers before both the overall share and the Fisher test are recomputed. The defensible reading is therefore: a physics prior does not demonstrably buy validation rigor — and settling the question requires the standardized protocols proposed in Section IX, not more subset cross-tabs.

The data concentration can be stated as a two-pole fragmentation metric (Fig. 4). Across 2019–2026, 23 primary papers — 26% of the 88 primary-application papers — rest on the same three public benchmarks (Severson/MATR, NASA, CALCE), whose combined cell count is on the order of a few hundred; meanwhile, the single largest source overall is private laboratory cycling (36 papers) whose data are rarely released. The empirical base is therefore not one benchmark monoculture but two poles: a small set of shared benchmarks onto which comparative claims crowd, and many unshared laboratory datasets over which accuracy claims cannot be audited at all. On the shared benchmarks, dataset-specific failure modes and implicit test-set familiarity inflate apparent comparability; on the private datasets, reproducibility is unavailable in principle. Both faces of this pattern — a small, slowly broadening empirical base under a multiplying literature, and unauditable private data beside it — make two-pole data fragmentation the most reproducibility-relevant feature of the classified subset's data landscape; benchmark reuse itself is erratic year to year (1–7 papers) rather than tracking output, so no saturation trend is claimed.

IV. Machine Learning for Battery State Estimation

SOC estimation is the most mature BMS function, and the classified subset shows how machine learning entered it. The early pattern pairs learning with classical estimation: Gaussian process regression for pack-level SOC with native uncertainty quantification [31], and — most influentially — recurrent networks embedded in unscented Kalman filters (UKFs), where the LSTM supplies a learned observation model and the filter supplies the recursive correction [5]. A second generation tuned network structure itself: particle-swarm-optimized LSTM improved accuracy and noise robustness [32], and evolutionary-assisted CNN–BiLSTM hybrids report sub-1% drive-cycle errors [33]. Simulation-informed training pipelines substitute automotive-accurate simulated data for exhaustive cell testing [34], comparative benchmarks on standard drive cycles weigh ML regression families [35], pack heterogeneity is surveyed in a joint review of pack-SOC, cell-inconsistency, and balancing estimation [36], unified frameworks estimate SOH and SOC jointly from partial charging data [37], large-format applications extend estimation to containerized storage strings [38], and a review of cutting-edge SOC algorithms weighs accuracy against embedded compute budgets [39] alongside primary comparative benchmarks [40].

The most substantive recent shift is toward hybrid schemes in which a filter, an equivalent-circuit model, or a physics prior constrains the learner: a genetic algorithm (GA)-optimized LSTM coupled with improved adaptive Kalman filtering reaches low error under measurement noise that breaks pure-network estimators [41]; equivalent-circuit-informed ML on impedance features estimates SOC across temperatures [42]; and physics-informed recurrent

TABLE III
Cross-tabulation of fleet-data use against online validation (103 classified papers). Fleet-data use describes the primary data source; online validation describes the validation protocol; their overlap is 8 of 10 fleet-data papers. A laboratory-data online demonstration was recoded to experimental hardware after full-text re-check (per the codebook's absorbing rule, Box 1). The right-hand column pools the environment-only validation categories defined with the classification framework.

	Online / real-vehicle validation	All other validation protocols	Total papers
Real-vehicle / fleet data	8	2 (cross-domain holdout; case study)	10
Other data sources	0	93	93
Total	8	95	103

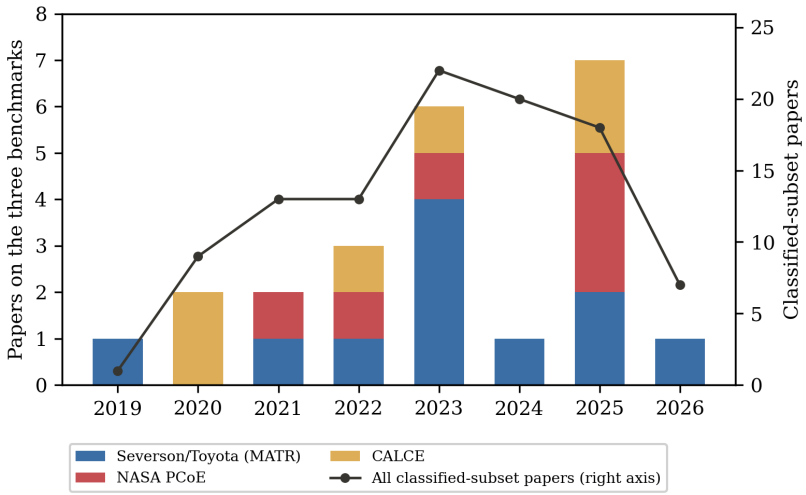


Fig. 4. Benchmark reuse versus classified-subset output. Papers in the classified subset using the three dominant public benchmarks per year (stacked bars) against classified-subset papers per year (line). Both quantities describe the curated 103-study subset, not the eligible corpus.

structures embed battery dynamics directly into the training objective [10]. These schemes trade engineering complexity for extrapolation behavior — the property that matters most onboard, though the corpus can verify only the premise of that argument: no subset study measures out-of-envelope degradation directly, and the validation audit (Section III-C) detects no difference in protocol stringency between hybrids and purely data-driven peers. Yet 11 of the 13 subset SOC primary papers rest on the authors' own laboratory cells and one on laboratory impedance measurements, 9 of the 13 on a single cathode chemistry (NMC), and the thirteenth is a simulation-based EV study [34]; the cluster's only operation-scale demonstration, a stationary containerized storage system rather than a vehicle [38], is itself coded on laboratory data under the released labels, with its re-reading as field-stationary data logged among the codebook patch candidates; benchmark heterogeneity makes cross-paper comparison nearly meaningless [1] [18]. Within the subset's own reporting, accuracy on laboratory cells no longer appears to be the binding constraint in this cluster; guaranteed accuracy on arbitrary packs over full cell aging, which the onboard estimator would need, is not demonstrated anywhere in the subset.

V. Machine Learning for Health Estimation and Prognostics

SOH estimation and RUL prediction form the largest functional block of the classified subset (32 primary papers: 19 SOH, 13 RUL). The founding pattern is classical: extract health-sensitive features from partial charge or discharge curves, then regress with Gaussian process or similar shallow models — incremental-capacity analysis with explicit uncertainty bounds [43], differential thermal voltammetry [44]; Gaussian-process classification of impedance spectra that identifies degradation modes [45]; and deep Gaussian processes coupling prognosis with degradation-mode identification [46]. Measurement relaxations followed: random partial charging data proved sufficient for accurate SOH [47], and charging-segment adjustment refined features for vehicle-level use [48]. The landmark systematization is Roman et al.'s 179-cell pipeline showing that relaxed charge-curve features support accurate SOH estimation across a heterogeneous commercial fleet [49].

Deep architectures entered as sequence models over capacity-fade trajectories and representation learners over raw curves: graph convolutional networks fusing multiple health indicators [7], graph networks over partial discharge curves [50], BiLSTM–Transformer hybrids under fast-charging stress [8], hybrid-fusion networks for real-time prediction streams [51], and interpretable deep learning with transparency mechanisms [52]. Field-data studies

absorb real-vehicle noise [6] and estimate pack-level SOH from fleet streams [53], while deep models estimate SOH without additional degradation experiments by learning trajectory features that generalize across chemistries [54]. Early-life prediction — forecasting lifetime from the first cycles — compresses test times from months to days [3] [55] [56], hybrid physics-plus-learned models give early RUL from limited cycles [57], the approach extends to lithium-metal chemistries [58], and it now runs on embedded boards in real time [59].

The prognostics cluster shows the clearest physics-informed turn: physics-informed online degradation models [60], hybrid reinforcement learning that reduces the data demand of SOH estimation [28], a generalizable PINN from limited early-cycle data [61], EIS-informed physics-constrained networks [62], a generic physics-informed RUL framework for small early-life samples [63], multi-feature hybrids across datasets [64], recurrent networks anticipating uncertain duty [65], and signal-decomposition hybrids on noisy capacity series [66]. Transfer learning attacks generalization directly: domain-adversarial training across operating conditions [11], deep transfer with partial charging curves across cell types [12], transfer under fast-charging protocols [67], cross-chemistry EIS estimation [68], and a general framework quantifying the laboratory-to-fleet domain gap [69]. The critical assessment: the cluster's accuracy claims are established overwhelmingly on public benchmark cells — 10 of the 32 primary studies in this complex use within-cell or k-fold validation (17 of the 88 primary papers in total) — and the transfer-learning papers, the honest exception, reveal how much shift remains unaddressed. SOH estimation under controlled conditions shows signs of approaching incremental returns within this subset; calibrated lifetime and safety margins for arbitrary packs in the field remain open [2] [20].

VI. Machine Learning for Charging Control and Optimization

Charging control is the function where machine learning has changed experimental practice most directly. The defining result is Attia et al.: a Bayesian optimization loop identified high-cycle-life protocols among 224 candidates in 16 days of closed-loop cycling, against more than 500 days for an exhaustive search without early prediction [4]. Ensemble surrogates make multi-objective strategy optimization tractable [71], and Bayesian fast-charging optimization reduces charge time under degradation constraints [78]. A second line embeds degradation mechanisms into the decision itself: machine-learning fast charging that perceives and regulates internal lithium-plating onset enables plating-free rapid protocols [70], adaptive multistage schemes adjust stage boundaries against measured degradation [79], module-level optimization manages temperature gradients [80], and pack-level extensions coordinate balance-aware charging [81] [82].

The deployment gap is most concrete here: of the eight charging papers, four demonstrate on laboratory hardware — including the 224-candidate closed-loop campaign above — and the remaining four rest on simulation-level evidence (three simulation-only, one with hardware spot-checks). No subset charging paper demonstrates a learned strategy validated online on a vehicle. The unsolved problem is certifiable learning-based charging — which is why mechanism-aware schemes such as plating-perception control [70] are the cluster's most deployment-relevant results.

VII. Machine Learning for Thermal Management, Faults and Safety

Thermal management (7 primary papers) divides by purpose. For estimation, ANN models predict cell temperature across operating conditions [72] and internal-temperature sensing extends to ultrahigh-rate discharge where surface measurement fails [83]. For design, ML surrogates amortize expensive computational fluid dynamics (CFD) evaluation across optimization loops [84] [73], with experimentally validated parallel-liquid-cooling designs closing the loop [85]. For control, ANN controllers demonstrate parasitic-load reductions in simulation [86] — but no subset paper closes this loop on hardware, leaving thermal control where charging control was five years ago. Physics-informed heat-generation estimation embeds thermal priors into the learning problem [9], and the cluster is consolidated by a state-of-the-art review [25].

Fault and safety applications (7 primary papers) divide into diagnosis of realized faults and prediction of impending hazards: grid-search-optimized SVMs classify sensor and connection faults [74], curvilinear-distance methods detect multiple pack fault types in operating fleets [87], and multi-scenario deep anomaly detection covers real EV operating envelopes [88]; force-signal deep learning catches lithium-plating-type defects at manufacturing [89]. The hazard-prediction side is increasingly model-based: multiphysics models coupled with ML predict module-level thermal-runaway progression [75], a multiphysics-informed DeepONet augments scarce experimental events with virtual data [90], and multi-source domain transfer moves thermal-runaway diagnosis across fleets with small target samples [76]. The reviews in this cluster are candid that laboratory fault diagnosis has historically failed to transfer to real vehicles [24] [23]; the fleet-data anomaly-detection studies are therefore the cluster's most valuable trend, even though label scarcity for rare faults keeps evaluation largely qualitative.

VIII. Cross-Cutting Directions

A. Physics-informed learning, interpretability, and uncertainty

Physics-informed and hybrid model–data learning is the classified subset's most consistently rising methodological theme (22 primary-application papers across all functions; Table I) — in state estimation [10] [91], heat-generation estimation [9], prognostics [60], and impedance-based diagnostics — consolidated by a physics-informed

TABLE IV
Representative classified-subset studies by management function and method family. Entries gloss each study as: study summary (data basis; headline contribution). References in brackets refer to the numbered list.

Function	Method family	Representative studies (data; contribution)
State estimation (SOC)	Filter-embedded learning; meta-heuristic DL	LSTM+UKF embedded estimator (laboratory cells; learned observation model in a UKF) [5]; PSO-LSTM (laboratory cells; meta-heuristically tuned recurrent SOC) [32]; GA-LSTM with adaptive Kalman filtering (laboratory cells; low error under noise) [41]
Health estimation (SOH)	Classical ML; deep learning; transfer learning	ML pipeline over 179 cells (own cell fleet; relaxed charge-curve features) [49]; deep learning without extra degradation experiments (public MATR dataset; cross-chemistry generalization) [54]; real-world vehicle SOH (fleet data; field-noise robustness) [6]
Prognostics (RUL)	Classical ML; physics-informed; RL	Early-life cycle prediction (public MATR dataset; about 9% median test error) [3]; physics-informed early-cycle RUL (multiple datasets; small-data prognosis) [63]; deep-RL prognostic model fusion (public NASA dataset; adaptive model selection) [13]
Charging control	Bayesian optimization; mechanism-aware control	Closed-loop Bayesian protocol optimization (hardware prototype; 224 candidates in 16 days vs >500 days of exhaustive search) [4]; plating-perceiving fast charging (hardware prototype; plating-free protocols) [70]; ensemble-surrogate charging optimization (laboratory cells; multi-objective strategies) [71]
Thermal management	artificial neural network (ANN) surrogate; physics-informed; optimization	ANN temperature prediction (laboratory cells; low-error thermal model) [72]; PINN heat-generation estimation (simulation; embedded thermal physics) [9]; ML-assisted battery thermal management system (BTMS) design (simulation; multi-objective cooling design) [73]
Fault & safety	Shallow ML; multiphysics + ML; transfer learning	SVM fault diagnosis (hardware prototype; sensor and connection faults) [74]; multiphysics-ML runaway prediction (simulation and hardware; module-level hazard) [75]; cross-fleet runaway-diagnosis transfer (fleet data; small-sample diagnosis) [76]
Cross-cutting	Physics-informed; explainable AI (XAI); digital twin; LLM	EIS diagnostics PINN review (multiple studies; paradigm synthesis) [21]; uncertainty-aware early-life prediction (public MATR dataset; calibrated forecasts) [77]; LLM-based SOH estimation with uncertainty-aware fallback (field data; subset's earliest LLM estimator — an earlier journal study was missed by the retrieval; see the foundation-models section) [15]

EIS-diagnostics review [21] — with physics-constrained autoencoders improving health prediction over purely learned counterparts [92] and a reviewed physical-plus-data-driven fusion literature for internal-state estimation [93]. The caution is that most physics-informed studies embed simplified physics — equivalent-circuit or reduced electrochemical models — so the hybrid inherits model bias even as it gains structure, and its validation is no stronger than that of purely data-driven peers (Section III-C). A smaller but consequential cluster makes learning systems inspectable: explainability-driven feature selection improves estimator robustness [94], interpretable ML links manufacturing coating parameters to cell capacity [95], interpretable hybrid ML untangles degradation chemistries in lithium–sulfur cells [96], and uncertainty-aware models deliver calibrated early-life predictions [77]. For safety cases only calibrated uncertainty has a route into certification; attribution supports debugging.

B. Digital twins, cloud BMS, and new sensing

System-level architecture is emerging as a research object: digital twins track individual battery aging through incremental learning [97] and couple LSTM surrogates with real-time thermal observation [98]; cloud platforms perform in-situ life prediction at fleet scale [99]; second-life workflows use impedance-based accelerated SOH estimation to triage retired packs [100], and learned battery models with numerical search estimate real-time state-of-power [101]. Non-electrical sensing broadens the measurement base — ultrasonic signatures provide a SOC channel orthogonal to voltage sensing [102] — and ML mappings from equivalent-circuit parameters to electrochemical properties give classical BMS models a physically interpretable learned layer [103]. These settings differ in latency budgets, error tolerance, and update cadence, and the field is beginning to design for those differences. Production-side mapping review [104], coating interpretability [95], and defect detection [89] complete the picture: cell quality is itself a BMS input.

C. Foundation models and LLM-based management

The classified subset's youngest theme is generative, and it arrives with a prior-art caveat: LLM-based battery state estimation already exists beyond this corpus: a 2025 IEEE Transactions on Energy Conversion letter [29] that this review's Crossref retrieval failed to return — a recall

miss, not a venue exclusion — so no study in this corpus can claim to be the first of its kind. Within the subset the published studies are of two distinct kinds: a domain-specific LLM for risk analysis of battery energy-storage-system incidents — station-side rather than onboard technology — [14], and HA-LLM, which crosses the estimation boundary by using field vehicle data to estimate SOH with uncertainty-aware control, coordinating vehicle and cloud [15]. The status of [15] should be stated precisely for what follows: LLM-based estimation is no longer hypothetical — what remains normative is the claim that before any LLM-based estimator feeds actuation, it must publish calibrated intervals, structured outputs, and a latency budget, and it must not hold direct cell-level protection authority; that role belongs to physics-based envelopes and calibrated numeric estimators. Table V makes this division of labor explicit; its right-hand column is guidance proposed here, not claims about the published studies.

IX. Research Gaps, Reporting Standards, and Future Directions

Cross-referencing the bibliometric analysis of Section III with the function-level reviews yields six structural gaps. (1) Validation practice: 17 of the 88 primary papers use within-cell or unsupervised k-fold validation, and only 8 of the 88 primary papers (every online demonstration in the subset is a primary application) evaluate online or on fleets; standardized cross-cell, cross-chemistry, and lab-to-fleet transfer protocols should become reporting norms, as already practiced by the transfer-learning and general-framework studies [69] [68]. (2) Two-pole data fragmentation: the three public benchmarks sustain 23 primary papers and 36 rest on private laboratory cells that are rarely released; federated learning over fleets, anonymized fleet-data releases, and multi-institution cycling campaigns — in the spirit of the 179-cell pipeline [49] — would diversify the evidence base, and papers should at minimum disclose split definitions, seeds, and compute budgets. (3) Edge deployment: only isolated works treat onboard compute, memory, and latency as first-class constraints [59] [105]; quantization-aware training, resource-constrained architecture search, and cloud–edge hybrid designs are the natural research program. (4) Trust: uncertainty must be calibrated rather than post-hoc [77], interpretability must bound behavior rather than merely attribute it, and hybrid physics–data architectures are the most credible certification path because part of their behavior is analytically specified. (5) Foundation-model integration: an LLM-based estimator [15] and the risk-analysis LLM [14] suggest a division of labor — foundation models for reasoning and orchestration, conventional ML for calibrated estimation, physics models for safety envelopes — that should be formalized with grounding and latency guarantees. (6) Under-served functions: balancing is nearly untouched beyond pack-level insight [30], charging needs certifiable adaptive protocols, and thermal management remains simulation-dominated [25]; these are control problems where errors act on the battery, so closing them requires the safety-case machinery of gap 4 as much as new algorithms. Three follow-on subsections turn these gaps outward — to what the automotive safety standards demand of these functions (IX-A), how this review is positioned relative to prior reviews (IX-B), and what the next research cycle should produce (IX-C).

A. From corpus gaps to safety-standard requirements

The standards invoked in Section I — UN GTR No. 20 (electric-vehicle safety, established in the Global Registry in 2018 [106]; Amendment 1, submitted by the European Commission on behalf of the Informal Working Group on Electric Vehicle Safety as formal document ECE/TRANS/WP.29/GRSP/2026/4 (status report ECE/TRANS/WP.29/GRSP/2026/3) [107], considered at the 79th GRSP session in June 2026 and, at this review's 18 October 2026 finalization, pending before WP.29 with adoption listed for its 17–20 November 2026 session; it adds thermal-propagation provisions with two compliance options — a physical propagation test or a documented risk-management approach —), GB 38031 (GB 38031-2020, now superseded, and the strengthened GB 38031-2025 [108], effective July 2026, which replaces the fire/explosion-warning criterion with no-fire/no-explosion thermal-diffusion requirements and adds bottom-impact and fast-charge-cycle tests), ISO 26262:2018 (functional safety of road-vehicle E/E systems [109]), ISO/PAS 8800:2024 (safety and artificial intelligence — the road-vehicle extension of ISO 26262 and ISO 21448 to learned components [110]), and ISO 21448 (safety of the intended functionality, SOTIF [111]) — do not prescribe estimation algorithms, but they define what a deployed estimator or controller must be able to evidence, and Table VI translates the subset's gaps into those terms. The mapping is this review's synthesis at requirement level rather than clause level, and it is stated so that it can be checked against the standards texts themselves: the point is not legal advice but that a validation protocol reported in a paper corresponds, or does not, to evidence a safety case can use. The inference rule behind every "supports a standards argument" statement in this section is explicit: a paper's evidence can support such an argument only if it demonstrates performance on at least one held-out integration unit (a cross-cell holdout or stronger isolation protocol) and reports the environment in which that evidence was obtained; by construction, within-cell and k-fold evidence cannot support the argument, because the safety case is about unseen cells and unseen duty. Three consequences stand out for automotive deployment. Within-cell and k-fold validation (17 primary papers) cannot support an ISO 26262 argument, because the safety case is about unseen cells and unseen duty and the protocol never tests an unseen unit; a standards argument must therefore be argued from the isolation evidence, not asserted from headline accuracy. Thermal-propagation early-warning latency is a timing requirement, so fault-detection papers evaluated on curated laboratory windows

TABLE V
Safety and verification taxonomy for foundation-model components in the BMS stack. The right-hand column is normative guidance proposed by this review, not claims about the published studies.

Dimension	Classical / deep ML estimators (established practice)	Published LLM studies (2026)	Required guardrail (this review)
Role in the stack	Calibrated numeric estimation and detection	Stationary-BESS incident risk analysis [14]; LLM-based SOH estimation with uncertainty-aware control on field vehicle data [15]	Estimation should belong to calibrated ML; the LLM SOH estimator of [15] is the earliest in the classified subset — an earlier journal-published LLM estimator, missed by this review's retrieval, is [29] (the foundation-models section) — and must show calibrated intervals before any actuation-adjacent role
Grounding	Trained on domain cycling data	Domain-tuned corpora and knowledge-enhanced prompting	Retrieval over verified corpora with source citations; no ungrounded recall
Dominant failure mode	Bounded numeric error under distribution shift	Open-ended hallucinated text	Outputs restricted to structured, checkable forms; never free text into protection logic
Uncertainty	Gaussian-process / conformal intervals in a subset of studies	Not reported for incident risk analysis [14]; Monte-Carlo uncertainty estimates with selective physics-estimator substitution demonstrated in the SOH estimator [15]	Calibrated intervals mandatory before any actuation-adjacent role
Latency and compute	Millisecond-class, onboard	Second-class, cloud-dependent	Offline analysis and planning only unless real-time evidence is shown
Safety authority	Bounded by physics envelopes and thresholds	Undefined in published studies	LLM components must never hold direct cell-level protection authority

under-report both the false-alarm rate and the detection-delay distribution the GB 38031-2025 / GTR-20 escape window needs. And charging or thermal control strategies validated only in simulation lack the hardware-in-the-loop evidence base an automotive release process expects. The reporting checklist below (Table VIII) is deliberately aligned with this mapping — reporting an isolation argument and a validation environment is the minimum on which any standards conversation can start.

B. Position relative to prior reviews

The closest in scope among recent reviews is the performance-centric review of Prasanna et al. [27], which examines 240 peer-reviewed publications (from 361 initially retrieved records) spanning the whole EV powertrain — battery management alongside charging systems, motor control, and energy-consumption prediction — and aggregates reported performance metrics (root-mean-square error, RMSE, and R^2) into a quantitative benchmarking of algorithm families, concluding that hybrid deep-learning architectures (LSTM-DDQN, CNN-BiGRU, transformer-based frameworks) consistently achieve superior accuracy ($R^2 > 0.95$, RMSE < 0.05). This review differs from that work in scope, unit of analysis, and question asked. In scope, it is narrower and deeper — battery-management functions only — which permits function-level cuts such as the standards mapping of Section IX-A and the per-function validation audit. In unit of analysis, it is protocol-centric rather than performance-centric: instead of pooling reported accuracy figures, it codes how each study's validation protocol isolates (or fails to isolate) its generalization claim. The two approaches are complementary, and the complementarity is pointed: an aggregation of self-reported RMSE and R^2 may inherit, unmeasured, the validation weaknesses this review quantifies within its own subset — there, 17 of 88 primary papers validate on information-leaking splits, and headline accuracies concentrate on three shared benchmarks — so where Prasanna et al. answer which algorithm family performs best as reported, this review asks how much the reports can be trusted, and what a study must disclose for its claims to be checkable. Adjacent cross-functional surveys exist for degradation and estimation methods; Table VII states the comparison axis by axis across all of them: a two-tier corpus with record-level screening provenance, a within-corpus measurement of validation protocol strength across functions, a standards-grounded gap analysis, and a released, re-run-able analysis pipeline. Where prior reviews ask "which method works best for function X," this review asks "which claims can actually be checked — and what would it cost the reader to trust them," which is the question a scoping review with a curated classified subset is positioned to answer, and the question that performance-centric aggregation cannot ask of itself.

C. Outlook: what the next research cycle should produce

Concretely, and tied to the gaps above, the highest-value near-term work falls in six directions. (a) Duty-cycle-aware validation: public cross-cell benchmarks that pair cycling data with recorded EV drive profiles, so that cross-chemistry and cross-domain generalization can be tested against transportation duty rather than laboratory schedules; (b) federated fleet evidence: privacy-preserving SOH/SOC studies over live fleets (≈2% of the eligible corpus today), with the 179-cell pipeline as the template; (c) benchmark diversification: multi-institution

TABLE VI
Mapping BMS functions to requirements in UN GTR 20 (Amendment 1 thermal-propagation provisions), GB 38031-2020/2025, ISO 26262:2018 with ISO/PAS 8800:2024, and ISO 21448 (SOTIF): what deployed ML components must be able to evidence. Regulatory statuses are stated as of 18 October 2026, and a 'strong protocol' in the subset-status column follows the protocol-only definition of Section II-C (cross-cell, cross-chemistry, or cross-domain holdout, or online/real-vehicle demonstration). The mapping is this review's synthesis at requirement level (not clause-level legal citation); "subset status" gives the corresponding classified-subset primary-paper counts from Section III-C.

BMS function	What the standard demands of a deployed function	Current subset status
SOC estimation	ISO 26262:2018 treats estimation error that can enable overcharge/overdischarge as an E/E-system safety hazard to be handled through the safety lifecycle; ISO/PAS 8800:2024 extends this to learned elements (assurance argument over data, model, and known insufficiencies); UN GTR 20 / GB 38031-2025 treat overcharge as an abuse-protection requirement	3 of 13 SOC primary papers use a strong isolation protocol; none validate on real-vehicle data; 12 rest on laboratory-scale data
SOH / RUL prognosis	Capacity and power fade margins underpin thermal-propagation early warning and end-of-service decisions; degradation assumptions must hold across chemistry, duty, and environment (ISO 26262 hazard analysis; ISO 21448 for performance limitations)	20 of 32 SOH/RUL primary papers use a strong protocol; 4 use real-vehicle data; accuracy claims rest mostly on three public benchmarks (16 of the 32 SOH/RUL primary papers; 23 across the whole subset)
Fault detection	Thermal-propagation containment and early warning must both be evidenced: GB 38031-2025 requires that after single-cell thermal runaway the pack or system neither fire nor explode over a two-hour observation window, that a thermal-event alarm still be issued, and that smoke not harm occupants — superseding GB 38031-2020's five-minute pre-fire alarm; the GTR 20 Amendment 1 proposal (ECE/TRANS/WP.29/GRSP/2026/4) carries a $\geq$5-min advance warning before cabin hazards, with test or risk-management compliance options; coverage must be argued and audited (ISO/PAS 8800 assurance argument)	2 of 7 fault/safety primary papers validate online on fleet data; detection-latency and false-alarm distributions are rarely reported
Charging control	Overcharge, over-temperature, and thermal-cycling limits are hard constraints under GB 38031-2020/2025; adaptive strategies must be bounded by protection envelopes and argued in the safety case	0 of 8 charging primary papers use a strong protocol; 4 demonstrate on laboratory hardware, none online
Thermal management	Temperature-gradient and propagation-containment requirements after fast-charge or abuse (GB 38031-2025 thermal diffusion, bottom-impact, fast-charge-cycle tests); the cooling function must be verified on hardware	0 of 7 thermal primary papers use a strong protocol; 2 exercise laboratory hardware; no learned control loop is validated on hardware

TABLE VII
Position relative to prior reviews. "Prasanna 2026" denotes a 2026 performance-centric ML review of EV powertrains with BMS and charging scope. The function-specific and method-specific reviews named in Sections I–II cover single functions or methods, so none of them, to our knowledge, addresses the axes below; here the comparison is per axis, not per review. This review differs most in the validation measurement (within a curated subset), standards linkage, and released record-level artifacts.

Axis	This review	Prior cross-functional / powertrain reviews
Corpus	Two-tier: 799 rule-based eligible records + 103 classified subset	Fixed study lists; no record-level selection provenance in the works we examined
Validation analysis	Protocol classes defined, per-paper assignments released, method-family contrast tested with sensitivity ranges; within-corpus measurement, not a field census	Narrative remarks on data scarcity; protocol strength not quantified, to our knowledge
Standards linkage	Function-to-requirement mapping vs. UN GTR 20, GB 38031-2020/2025, ISO 26262:2018, ISO/PAS 8800:2024	Not addressed or addressed qualitatively
Recommendations	Twelve-item reporting checklist (BMS-ML-RC) + six research directions	General future-work statements
Reproducibility	Released papers.csv, screening provenance, codebook, and statistics scripts	No released record-level dataset that we could locate

cycling campaigns two orders of magnitude larger than MATR/NASA/CALCE and released with DOI-backed artifacts, addressing the two-pole fragmentation of Section III-C; (d) pack- and vehicle-level ML: estimation validated at the integration level where UN GTR 20 and GB 38031 requirements bite, instead of cell-level evidence transplanted upward; (e) functionally safe learning: ML estimators whose uncertainty handling and assurance arguments are specified against ISO 26262 safety goals and ISO/PAS 8800, with ISO 21448/SOTIF reasoning for capability limitations, making certifiable charging and thermal control reachable; and (f) honest foundation-model integration: retrieval-grounded LLM agents with published latency and calibration budgets, capable of the coordination role while estimation authority stays with calibrated numeric models — extending, not contradicting, the corpus's LLM estimator [15]. A seventh, methodological direction concerns the evidence base itself: a v2 re-retrieval of this corpus with cursor paging — the upgrade the retrieval-depth pilot of Section II-A measures

as this pipeline's single highest-leverage methodological step, worth on the order of 5,000 additional eligible records — so that the next synthesis can measure field-level shares over a registry-deep sample rather than a top-20-per-pass cap. A guardrail on all seven directions: the BMS-ML-RC checklist below is designed to be attached to a paper at submission, so that the next generation of BMS-ML studies arrives auditable by construction.

D. A reporting checklist for machine-learning BMS studies

The most concrete artifact this review can offer is procedural. Modeled on the CONSORT and PRISMA checklists, Table VIII proposes a twelve-item BMS-ML reporting checklist (BMS-ML-RC): the minimum metadata a study should disclose for its accuracy claims to be comparable, reproducible, and poolable — and, per Section IX-A, admissible in a safety discussion. Its derivation is explicit: items 4, 5, and 11 operationalize the validation-protocol classes defined in Section II-C and measured in the classified subset; items 1–3 and 6–10 respond to the reporting omissions observed across Sections III–VIII; item 12 is the artifact-availability norm this paper itself follows. What the checklist is and is not: it is a proposal grounded in the classified subset's measured weaknesses, not a community-validated instrument — no Delphi or expert-consensus process was run for it, only the validation-protocol, data-source, and baseline-comparison items are score-measured in this corpus, and it has not been applied as an audit to all 103 studies. The review's own reporting follows the PRISMA-ScR flow structure (Fig. 2), but the protocol was not prospectively registered; where this section says a practice should be reported, that is a recommendation supported by the evidence above, and we suggest journals consider it as a submission checklist.

TABLE VIII
Proposed BMS-ML reporting checklist (BMS-ML-RC), modeled on CONSORT/PRISMA reporting standards. Items 4, 5, and 11 derive from the validation-protocol classes defined in Section II-C and measured in the classified subset; item 12 is the artifact-availability norm this paper itself follows. The checklist as a whole has not been score-audited across all 103 classified papers — it is a proposal, not a survey result.

#	Reporting item — what must be reported
1	Dataset identity — exact name and version of every dataset (e.g., Severson-MATR, NASA, CALCE); download source
2	Cell population — number of cells, chemistry, manufacturer, format, and batch for every training and test set
3	Sensor and sampling context — measured channels, sampling rate, sensor class, temperature conditions
4	Split protocol — split type (within-cell / cross-cell / cross-chemistry / cross-domain) and isolation argument
5	Out-of-distribution evaluation — at least one protocol where test conditions differ from training
6	Baselines — at least one classical baseline alongside the proposed architecture, on identical data
7	Randomness control — seeds or repeated-run statistics for every stochastic result
8	Uncertainty — calibrated intervals or error bars for headline claims, not point estimates alone
9	Compute budget — training compute and inference latency/memory on the target platform
10	Failure analysis — at least one documented failure mode and its distribution
11	Validation environment — simulation, laboratory cycling, hardware prototype, or online/fleet
12	Artifact availability — data and code deposit (DOI-backed) or an explicit stated exemption

X. Conclusion

This review has examined machine learning in battery management on a two-tier evidence base: a rule-based eligible corpus of 799 records from which field-level shares are measured within a documented capped retrieval, and an in-depth classified subset of 103 studies through which method, data, and validation practice are examined with a two-round 30-paper re-coding protocol — blind abstract-level round: three of five fields $\kappa \geq 0.88$; revised-codebook rule-application round on a shared evidence base: $\kappa = 0.92$–1.00 — and label-noise sensitivity ranges. Three conclusions stand out. First, and most critically, the proliferation of published accuracy claims is not matched by validation progress: only 8 of the 88 primary papers (8 of the 103 classified studies) demonstrate online or fleet operation, 17 of 88 primary papers use isolation-weakened splits, and 23 rest on three public benchmarks while 42 rest on private laboratory data (36 own-cell, 6 laboratory impedance); the physics-informed validation contrast is directionally consistent under both reported definitions, statistically unestablished and underpowered (protocol-only 36% versus 52%, $p = 0.32$; including-hardware 59% versus 74%, $p = 0.19$), so it is reported as no detectable difference, with meaningful differences in either direction not excluded — an open hypothesis rather than a settled result. Second, the retrieval's center of gravity is estimation at cell level: the SOC–SOH–prognostics query-theme union tags 366 of 799 records (46%) — a property of the capped retrieval and its query families, not a field-level share — whereas vehicle- and pack-level practice remains thin — precisely the level at which UN GTR 20, GB 38031, and ISO 26262 pose their requirements. Third, within the classified subset, method families have shifted from classical feature-based learners to deep sequence models and now physics-informed hybrids, with LLM-based studies appearing from 2026 — though LLM-based battery estimation began in other venues in 2025; this is a within-subset sequence that the eligible pool's method-theme tags do not mirror, so no field-level family rise is measurable at either tier. The near-term agenda follows directly: standardized cross-domain validation, diversified fleet data, pack- and vehicle-level ML aligned with the standards mapping of Section IX, calibrated uncertainty, and a strict division of labor between foundation models, calibrated estimators, and physics-based safety envelopes. If the last five years were about discovering what machine learning can estimate about a battery, the next five will be decided by what it can be trusted to do — onboard, at fleet scale, across chemistries, and over full cell lifetimes.

Data and Code Availability

The complete evidentiary record of this review is released as a Zenodo deposit (DOI: https://doi.org/10.5281/zenodo.23436202): the classified corpus of 103 studies (papers.csv) with per-paper citation counts, the 799-record eligible corpus and its record-level screening provenance (screening_log.csv, all 1,550 screening decisions plus five forced-seed entries), the released title-gate and tagging term lists (tagging_terms.json), the per-paper validation-protocol audit (protocol_audit.csv), the classification codebook, the blind 30-paper inter-coder reliability study (locked coding sheets, agreement report, itemized disagreement table), and the label-noise sensitivity analysis over headline counts. The complete search, screening, verification, and figure-generation scripts ship inside the same deposit and regenerate every figure and count in Section III; a development-form mirror is kept on GitHub.

Acknowledgment and Declaration on AI Tools

This work received no specific funding, and the authors declare no competing interests. Author contributions: H. T. Siddiquee conceived the study, executed the search, screening, and verification pipeline, performed the full-text classification, coded the released labels, and drafted the manuscript; S. S. I. Ariya and J. I. Chowdhury performed the blind 30-paper re-coding of the reliability study (coders B and C, Section II-C) before any verification work, and contributed to corpus curation, verification of classifications and counts, and manuscript revision. During the preparation of this manuscript the authors used the following tools, each for a stated purpose: an AI language model (GLM, Z.ai) assisted in drafting and adapting the literature-pooling, analysis, and figure-generation scripts; OpenAI ChatGPT, Grammarly, and QuillBot were used for language editing (grammar, readability, and sentence restructuring). No AI tool decided study inclusion, and no AI tool assigned or altered any released classification label: all screening decisions and all released labels were made by the authors; after the label set was frozen, the AI research assistant (coder D) independently re-coded the validation field as a disclosed robustness audit (Section III-C), and no AI-assigned code altered any released label. Section II-C documents who coded what, and the reliability files release the coding evidence. All included references were verified against the Crossref registry by direct DOI resolution, all counts and figures were regenerated programmatically from the released dataset, and the authors reviewed and take full responsibility for the content of the published article.

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